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Research paper

An algorithm for dip point detection in lithium–sulfur battery cells

Zahra Nozarijouybari, Catherine Fang, Mahsa Doosthosseini, Chu Xu, Hosam K. Fathy *

Department of Mechanical Engineering, University of Maryland, College Park, United States of America

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ABSTRACT

This article examines the problem of developing a simple, model-free algorithm for detecting and identifying the time instant when a Lithium–Sulfur (Li–S) cell passes through its "dip point" during discharge. The dip point marks a sharp transition between two different sets of redox reactions involved in Li–S battery discharge, and is characterized by a significant change in the slope of battery potential with respect to charge processed. This makes it possible to detect the dip point accurately using a simple algorithm that uses a moving-horizon least-squares method to estimate the above slope, then detects the dip point by detecting changes in this slope. We validate this algorithm both using a physics-based battery simulation and experimentally, using custom-fabricated Li–S coin cells. One potential benefit of this algorithm is the degree to which it makes it possible to pinpoint battery cell arrival into the low plateau region accurately, which is important in light of the well-recognized difficulties associated with Li–S battery state estimation in this region. This opens the door to potential innovations in Li–S battery pack balancing that rely on dip point detection as a simpler and reliable alternative to full battery state estimation.

1. Introduction

This article presents an algorithm for detecting the passage of a lithium–sulfur (Li–S) battery cell through a specific feature of its discharge voltage curve, known as the "dip point". The article's motivation stems from both the appeal of the Li–S battery chemistry and the need for battery management systems tailored for this chemistry. Li–S batteries have attracted significant attention in recent years thanks to at least four key advantages compared to more traditional lithium-ion (Li-ion) batteries. First, the Li–S chemistry offers a high theoretical specific capacity limit of 1672 Ah/kg and specific energy limit of 2600 Wh/kg, respectively [1]. Even the currently achievable Li–S battery specific energy levels of 600 Wh/kg or more [2] are beyond the practical limit for Li-ion batteries (roughly 500 Wh/kg) [1]. Second, Li–S batteries have the advantage of operating in a range of temperatures from −40 to 60 °C [3], which makes them potentially attractive for low temperature applications. Third, from a safety perspective, Li–S batteries are less susceptible to catastrophic failures compared to Li-ion batteries. This is due to their intrinsic overcharge protection [4] caused by precipitation and the internal charge shuttle effect. Fourth, the fact that Li–S cathode materials consist mostly of abundant elements like sulfur makes the Li–S chemistry an inexpensive and environmentally friendly candidate for next-generation batteries.

As shown in Fig. 1, the main Li–S battery cell components include a lithium metal anode, a porous separator soaked with a liquid electrolyte, and a composite sulfur–carbon cathode. Electrochemical phenomena, including anode-side lithium dendrite formation and polysulfide shuttle between the cathode and anode, are responsible for many of the limitations of the Li–S chemistry. These limitations include low cycle life, low coulombic efficiency, high self-discharge, relatively low operating voltages, and the large volume changes associated with the charging/discharging of sulfur electrodes [5]. Much of the recent research on Li–S batteries focuses on addressing these limitations through battery material design [6]. Examples of this research include: *(i)* the optimization of the cathode-side sulfur–carbon mix to improve sulfur utilization [7]; *(ii)* the use of metal oxide host materials for sulfur confinement [8]; *(iii)* the use of a carbon interlayer coated on the separator (facing the sulfur side) to provide a buffer that slows down the transport of polysulfides [9]; *(iv)* the use of electrolyte additives such as $LiNO_3$ to facilitate the formation of a protective anode-side solid-electrolyte interphase (SEI) layer and improve coulombic efficiency [10]; *(v)* the use of graphene-based hybrid anode structures to alleviate anode-side polysulfide contamination [11]; and *(vi)* the examination of battery degradation phenomena such as lithium pulverization, SEI layer accumulation, and cell impedance growth [5, 12].

* Corresponding author.
E-mail address: hfathy@umd.edu (H.K. Fathy).

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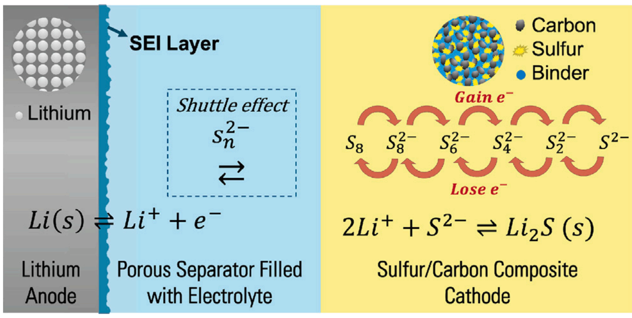

Fig. 1. Schematic of Li–S battery components with underlying electrochemical transformations.

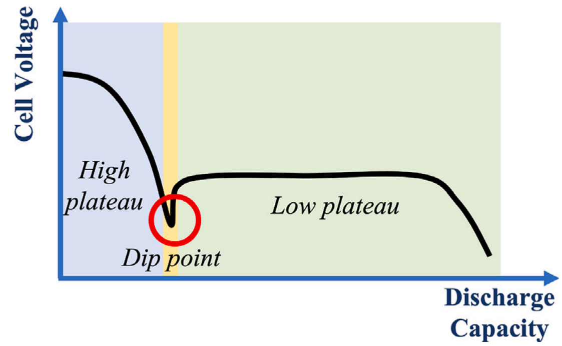

Fig. 2. A sketch of Li–S battery discharge voltage characteristics.

In addition to the above materials research, there is significant work in the literature on Li–S battery modeling, monitoring, and control. One objective of this research is to maximize Li–S battery pack performance and life through effective battery management, including pack balancing. Pack balancing is important for preventing battery overcharge or over-discharge. This, in turn, is important for maximizing performance and minimizing degradation even in relatively safe chemistries such as the Li–S chemistry.

Estimating battery state of charge (SOC) can be a valuable enabler for battery management functionalities such as pack balancing. The literature provides at least three different types of Li–S models that can be used for SOC estimation, with varying degrees of fidelity and complexity. First, there are simple equivalent circuit models (ECMs) that typically combine empirical open-circuit voltage curves with (potentially temperature- and charge-dependent) capacitor–resistor circuits [13,14]. Such ECMs have the potential to be used for online state of charge estimation, particularly in light of their simplicity [15,16]. Second, at the middle of the complexity range, there are zero-dimensional (0D) models which use the laws of electrochemistry to model underlying chemical reactions while neglecting species diffusion in Li–S batteries. This approach often leads to simpler and more computationally tractable models compared to the full modeling of battery diffusion–reaction mechanisms. The computational complexity of 0D models depends on specific choice of reactions to model. Examples of these choices include the simplified two-reaction and multi-reaction models proposed in the literature [17,18]. Third, the literature presents physics-based models that cover the redox reaction, precipitation, diffusion, and self-discharge dynamics in Li–S batteries. These models typically take the form of partial differential algebraic equations (PDAEs). They achieve higher levels of fidelity in describing the underlying battery physics, at the expense of more complexity and computational cost [19–21]. This complexity arises partly from the inclusion of numerous electrochemical (including precipitation) reactions, especially on the cathode side. One important reformulation of these models is the interpretation of PDAE battery dynamics using a "tank-in-series" formulation, enabling the simplification of these dynamics' complexity [22].

Regardless of the type of model used, model-based SOC estimation remains quite challenging for Li–S batteries, especially at low operating voltages. At least five different reasons exist for this difficulty. First, the multiplicity of species and reactions in Li–S batteries makes it difficult to define a single "state of charge", thereby motivating research on the estimation of the masses of different species in the battery instead [23]. Second, even if a simple aggregate SOC is desired, the presence of potentially significant self discharge means that common SOC estimation methods used for Li-ion batteries, including open-circuit voltage measurement and Coulomb counting, are ineffective for Li–S battery SOC estimation [16]. Third, significant observability issues exist at low voltages in Li–S batteries, making SOC estimation particularly difficult towards the end of battery discharge [23,24]. Fourth, the complex underlying reaction path with several dissolved species results in a non-monotonic relationship between battery voltage and discharge capacity, which makes SOC estimation hard. Fifth, Li–S battery model parameters, including achievable capacity, depend on the utilization of active sulfur in the cathode, which in turn varies with current density, temperature, and aging [16]. This makes it difficult to obtain accurate models of the underlying battery physics for the purpose of SOC estimation.

Compared to the above literature, the overarching goal of this article is to develop a model-free algorithm for detecting the passage of Li–S battery cells through a specific feature of their discharge voltage curves. The intent of this algorithm is not to estimate Li–S battery SOC, but rather to detect battery passage through a particularly important feature of its overall discharge curve. As sketched in Fig. 2, the discharge voltage curve for an Li–S battery with a liquid electrolyte typically exhibits three different features, namely: *(i)* a high plateau, *(ii)* a low plateau, and *(iii)* a transition point between these two plateaus. The high plateau is typically characterized by a monotonically decreasing terminal voltage, a low internal resistance, and a high self-discharge rate [16]. In this region, the active cathode material, S_8, receives electrons and forms polysulfides S_x^{2-}, where x can be 8, 6, or 4. In contrast, the low plateau is typically characterized with a nearly flat terminal voltage (assuming constant-current discharge) and limited power capability towards the end of discharge. In this region, further polysulfide reductions happen, where S_2^{2-} and S^{2-} are the dominant sulfide species. SOC observability is typically quite poor in this specific region, making SOC estimation difficult towards the end of discharge [23]. Finally, the transition point marks a fundamental change in the redox reactions occurring in the battery, and often takes the shape of a "dip point". The precise shape of this transition, including the "sharpness", of the dip point, is known in the literature to depend on a number of factors, including battery discharge rate [25]. In particular, research by both Andrei et al. [26] and Ren et al. [27] shows that the transition from the high to the low plateau regions is associated with the onset of Li_2S precipitation, which in turn requires some degree of electrolyte oversaturation. Therefore, the depth of the cell voltage drop during this transition can be modeled as a rate-dependent phenomenon, with higher C-rates typically associated with a more uniform particle size distribution, a more limited discharge capacity, and a less pronounced "dip".

To the best of our knowledge, this article is the first body of work in the literature presenting a simple, model-free algorithm for detecting the passage of Li–S battery cells through this transition point, or "dip point". One potentially valuable future application of this idea lies in the Li–S battery pack balancing domain, where the passage of different series-connected Li–S battery cells through this dip point at different time instants can be construed as a measure of pack imbalance. Such an approach is quite appealing compared to full-blown SOC estimation,

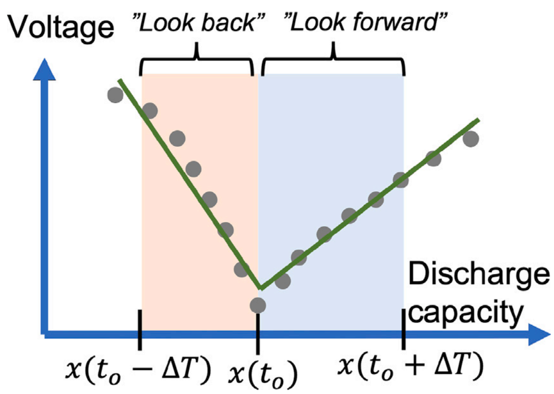

Fig. 3. Schematic of proposed regression approach.

thanks to its inherent simplicity. Moreover, as we show in this article, both in simulation and experimentally, detecting the transition between the high and low plateaus can be achieved accurately even at C-rates where this transition does not manifest itself as a "dip point". The fact that the proposed algorithm is model-free has the additional advantage of not being dependent on significant modeling and parameterization efforts for successful dip point detection.

The remainder of this article is organized as follows. Section 2 presents the proposed dip point detection algorithm. Section 3 and Section 4 apply this algorithm to both simulation-based and experimental discharge curves for laboratory-fabricated Li–S coin cells. Finally, Section 5 summarizes the article's findings and conclusions.

2. Dip point detection algorithm

Fig. 3 presents a simple schematic of the main idea behind the proposed dip point detection algorithm. At any given moment in time, t_o, the algorithm creates a "look-back" period and a "look-forward" period, separated from t_o by a user-selected time window ΔT. As shown in the figure, these two time periods do not necessarily correspond to equal amounts of battery discharge, in Coulombs. The main idea behind the proposed algorithm is to use linear least-squares regression to fit two straight-line relationships for measured battery terminal voltage versus discharge capacity, $x(t)$, during these two periods. The term "discharge capacity", in this context, is used in a manner consistent with the electrochemistry literature, in the sense that it refers to the total amount of charge removed from the battery, in Coulombs. The dip point is defined as occurring at the specific moment in time during discharge corresponding to the largest positive increase in the slope of battery voltage versus discharge capacity.

Intuitively, the intent of the algorithm is to search for the point where the battery discharge curve exhibits the largest positive curvature. This does not necessarily correspond to a discharge curve slope of zero, meaning that the algorithm does not search for the dip point by searching for a local minimum of the discharge curve. This is important because the transition between the high and low plateaus does not always manifest itself as a "dip point", especially at high discharge C-rates. Thus, our deliberate decision to define the "dip point" as the point exhibiting the largest positive change in discharge voltage slope versus discharge capacity renders the algorithm more robust to the various possible manifestations of the transition between the high and low voltage plateaus. Such robustness is important given the broad range of C-rates with which Li–S cells may be cycled in practical applications.

To state the proposed algorithm mathematically, let $y(t)$ and $u(t)$ be the measured battery output voltage and input discharge current, respectively, as functions of time. Moreover, let discharge capacity, $x(t)$, be given by:

$$x(t) = \int_0^t u(\tau) d\tau \tag{1}$$

At any given moment during battery discharge, battery output voltage, $y(t)$, depends on multiple factors, namely: the input current, $u(t)$, the discharge capacity, $x(t)$, battery state of health, battery temperature, battery material selection/design, and additional battery dynamics such as diffusion–reaction dynamics. To keep the dip point detection algorithm as simple as possible and minimize the need for modeling higher-order battery dynamics, one can potentially approximate battery output voltage solely as a function of $x(t)$ and $u(t)$, as shown below:

$$y(t) \approx f(x(t), u(t)) \tag{2}$$

In a **model-based** battery estimation algorithm, the form of the function $f(x(t), u(t))$ is often assumed to be known. In this **model-free** research, in contrast, the form of this function is not assumed to be known *a priori*. Instead, we use a first-order Taylor series expansion to approximate the function in the neighborhood of the present values of the battery discharge current, $u(t)$, and discharge capacity, $x(t)$.[1] Specifically, let $u_o = u(t_o)$ and $x_o = x(t_o)$. Then the function $f(x(t), u(t))$ can be approximated as follows:

$$f(x(t), u(t)) \approx f(x_o, u_o) + \gamma(x(t) - x_o) + R(u(t) - u_o) \tag{3}$$

In the above equation, the unknown constant γ represents the local slope of battery voltage with respect to discharge capacity, often referred to as the "dV/dQ" slope in the literature. Moreover, the constant R represents the effective series resistance of the battery. The proposed approach treats these two quantities as having distinct constant values during the "look-back" and "look-forward" periods, respectively. We use linear least squares regression to estimate the values of these constants during each of these two time windows. The dip point is then defined as occurring at that moment in time when γ experiences its largest positive change.

To illustrate the above regression process, consider the "look back" window for the special case of constant-current discharge at $u(t) = u_o$. Moreover, let δt represent the sampling time step used for data acquisition. Finally, denote an arbitrary moment in time during the "look back" window by $t_o - k\delta t$, where k is the number of sampling time steps taken backwards from t_o. In that situation, the above linear equation can be rewritten as follows:

$$y(t_o - k\delta t) \approx f(x_o, u_o) + \gamma(x(t_o - k\delta t) - x_o) \tag{4}$$

The above equation can be written as a standard linear regression equation of the form:

$$Y_k \approx \theta_1 + \theta_2 \phi_k, \tag{5}$$

where $Y_k = y(t_o - k\delta t)$ is the regression equation's output, $\theta_1 = f(x_o, u_o)$ and $\theta_2 = \gamma$ are the unknown linear regression parameters, and the corresponding regressors are 1 and $\phi_k = x(t_o - k\delta t) - x_o$, respectively. Based on this regression equation, one can assemble the following regressor and output matrices:

$$\vec{y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \quad \boldsymbol{\Phi} = \begin{bmatrix} 1 & \phi_1 \\ \vdots & \vdots \\ 1 & \phi_n \end{bmatrix} \tag{6}$$

where $n = \Delta T/\delta t$.

[1] The use of higher-order Taylor series expansions is possible, but the proposed dip point detection algorithm does not require such higher-order approximations, with their concomitant increase in the volume of data required for accurate estimation.

Given this regression problem formulation, one can finally estimate θ_1 and $\theta_2 = \gamma$ using the standard linear regression formula:

$$\hat{\vec{\theta}} = [[\mathbf{\Phi}]^T[\mathbf{\Phi}]]^{-1}[\mathbf{\Phi}]^T\vec{y} \tag{7}$$

The above regression formula furnishes two estimates of $\gamma = \theta_2$, corresponding to the two time periods preceding and following t_o. The difference between these two estimates of γ helps in detecting the dip point, the idea being that the dip point corresponds to the largest positive value of this difference. One implication of this simple proposed approach is that dip point detection occurs at least some time ΔT after the occurrence of the dip point, since a "look-forward" period of ΔT is required by the algorithm. This creates a tradeoff between small values of ΔT that may be more conducive to rapid detection versus large values that may potentially aid in more accurate detection by allowing the "look back" and "look forward" windows to be longer, and therefore to correspond to larger volumes of data. One final caveat of this proposed algorithm is the notion that the dip point corresponds to the maximum positive change in the estimate of γ. In a practical online battery management scenario, it is infeasible to wait until the very end of battery discharge to ascertain this maximum. Instead, the proposed algorithm declares that the dip point has been reached if the largest change seen in the value of γ since the beginning of discharge has not changed over some previous time window T_2. The implication is that the proposed algorithm will declare passage through the dip point some time duration of $T_2+\Delta T$ after the true occurrence of this dip point.

Implementing the above algorithm involves choosing values for the algorithm's parameters, such as the sampling time step, δt, the number of samples employed for regression, n, the time duration over which regression is performed, ΔT, and the optimization wait period, T_2. The values of these parameters represent fundamental tradeoffs between the algorithm's speed and accuracy. The remainder of this section provides some mathematical characterization of these tradeoffs for a representative dip point detection scenario, with the important caveat that the process used for this characterization is broadly generalizable to other scenarios. Specifically, consider the scenario where an Li–S cell is subjected to constant current discharge at some current $u(t) = I_o$. Moreover, suppose that the measurements of Li–S cell output voltage are corrupted by a zero-mean, independent, identically-distributed (i.e., white), and Gaussian voltage measurement noise process with variance σ_v^2 (in Volts squared). Under these assumptions, the regressor ϕ_k is given by:

$$\phi_k = x(t_o - k\delta t) - x(t_o) = -k\delta t I_o \tag{8}$$

Given this regressor, one can compute a Fisher information matrix that enables uncertainty quantification for the proposed estimator as follows:

$$\begin{aligned}[\mathbf{F}] &= \frac{1}{\sigma_v^2}[[\mathbf{\Phi}]^T[\mathbf{\Phi}]] \\ &= \frac{1}{\sigma_v^2}\begin{bmatrix} 1 & 1 & \dots & 1 \\ -\delta t I_o & -2\delta t I_o & \dots & -n\delta t I_o \end{bmatrix}\begin{bmatrix} 1 & -\delta t I_o \\ 1 & -2\delta t I_o \\ \dots & \dots \\ 1 & -n\delta t I_o \end{bmatrix} \\ &= \frac{1}{\sigma_v^2}\begin{bmatrix} n & -\delta t I_o \frac{n(n+1)}{2} \\ -\delta t I_o \frac{n(n+1)}{2} & (\delta t)^2 I_o^2 \frac{n(n+1)(2n+1)}{6} \end{bmatrix}\end{aligned} \tag{9}$$

The inverse of the above Fisher information matrix provides a Cramér–Rao lower bound on the best accuracy with which one can perform the proposed least squares regression. Moreover, the diagonals of this inverse provide the best-achievable variances of the two parameters estimated by the regression process. This means that the best achievable unbiased dV/dQ slope estimation variance, S^*, is given by:

$$\begin{aligned} S^* &= \frac{\sigma_v^2 n}{n(\delta t)^2 I_o^2 \frac{n(n+1)(2n+1)}{6} - (\delta t)^2 I_o^2 \frac{n^2(n+1)^2}{4}} \\ &= \frac{\sigma_v^2}{(\delta t)^2 I_o^2 n(n+1)}\left[\frac{1}{\frac{2n+1}{6} - \frac{(n+1)}{4}}\right] \\ &= \frac{12\sigma_v^2}{(\delta t)^2 I_o^2 n(n+1)(n-1)} \end{aligned} \tag{10}$$

The above expression captures the accuracy with which one can estimate the dV/dQ slope using the proposed algorithm once, either before or after the passage of the dip point. The two corresponding uncertainty distributions, under the above assumptions, are both zero-mean and Gaussian. The variances of these two distribution add up to provide the best variance, Σ^*, with which the *change* in the dV/dQ slope can be estimated:

$$\Sigma^* = 2S^* = \frac{24\sigma_v^2}{(\delta t)^2 I_o^2 n(n+1)(n-1)} \tag{11}$$

Several insights can be gleaned from the above expression, all of which make intuitive sense. First, the accuracy with which the above algorithm can estimate the change in slope at any given point in the battery cell's discharge becomes worse as the voltage measurement variance, σ_v^2, increases. Second, this accuracy improves as the amount of charge removed from the battery cell per sampling time period, $I_o\delta t$, increases. This amount of charge removed per sampling time period serves as a regressor in the estimation problem, with the important note that regression accuracy typically improves with increasing regressor magnitudes (provided the underlying regression model is accurate). Third, the larger the number of time steps used for regression, n, the higher the best achievable estimation accuracy. All of these conclusions point to fundamental tradeoffs in the design and implementation of the proposed algorithm. For instance, reducing voltage measurement noise variance improves the algorithm's performance, at the cost of a potentially more expensive sensor. Similarly, increasing the magnitude of the regressor, $\delta t I_o$, improves the accuracy with which the change in the dV/dQ slope can be estimated, but worsens the resolution of discharge capacity with which the dip point is detected, considering the fact that $\delta t I_o$ is the resolution with which changes in battery discharge are quantized. Finally, increasing the number of samples, n, used for regression provides an estimation accuracy benefit, but increases the volume of data required for dip point detection. This, in turn, increases the time lag between the passage of the dip point and the detection of this passage, which can be expressed as $n\delta t + T_2$. In summary, the above analysis provides a tool, grounded in Fisher information-based uncertainty quantification, for understanding and navigating the fundamental tradeoffs inherent in choosing the parameters of the proposed algorithm.

3. Experimental validation, Part 1: Cell fabrication

We validated the above dip point detection algorithm using both a physics-based battery simulation model and experimental laboratory test data. To perform the experimental portion of the overall validation effort, we fabricated 2016-type Li–S coin cells following the approach in [24,28]. The cathode consisted of approximately 56% sulfur powder (99.5% purity, Sigma-Aldrich), 24% Ketjenblack (EC-600JD, Nouryon), 10% carbon nanofiber (PR-24-XT-LHT, Pyrograf), and 10% polyvinylidene fluoride (PVDF, Sigma-Aldrich), by weight. The sulfur and Ketjenblack powders were hand-milled together for 15 min, then placed in a vacuum oven at 60 Celsius for 12 h. The carbon nanotubes were then added, and the solid cathode material mixture was hand-milled for 15 min. The PVDF was then dissolved into N-Methyl-2-Pyrrolidone (NMP, Sigma-Aldrich) using a magnetic stirrer, spinning at 450rpm, with a weight ratio of 5% PVDF to 95% NMP. The above materials were mixed using a magnetic stirrer for 24 h to form the cathode slurry. This slurry was coated onto a carbon-covered aluminum foil, with a coating thickness of approximately 220μm. The cathode sheet was then dried in a vacuum oven at 60 Celsius for 12 h. Cathode discs of 16 mm diameter were then cut, and placed in an Argon glove box (Vigor). Coin cells were then assembled from the above cathode discs, lithium anode chips (16 mm diameter and 0.6 mm thickness, MTI), a single porous polypropylene membrane (type 2400, Cellgard), plus 1–2 spacers and 2 drops of electrolyte per cell. The electrolyte used was 1M LiTFSI (Gotion) and 0.2M lithium nitrate (99.99% purity, Sigma-Aldrich) in a 1:1 DOL/DME mixture by volume.

A constant-current, constant-voltage (CCCV) charge/discharge protocol was used for generating experimental data from the above Li–S coin cells. A typical CCCV cycle involved charging/discharging these cells using a 0.1mA constant current, up to a maximum voltage of 3 V and a minimum voltage of 1.8 V, with a terminal CV current of 0.01 mA. Terminal voltage was measured with a sampling time of $\delta t = 5$ s, and the resulting battery discharge curves were used for algorithm validation, as shown in the figures. Moreover, a value of $\Delta T = 1000$ s was used in applying the dip point detection algorithm to the experimental cell cycling data.

4. Experimental validation, Part 2: Dip point detection results

Four different studies were performed to assess the proposed dip point detection algorithm: a simulation-based study, plus three distinct experimental studies. Fig. 4 illustrates the performance of the dip point detection algorithm in simulation. A zero-dimensional, physics-based Li–S battery model, developed in earlier research by the authors [24], was used for simulating Li–S cell discharge. For the ideal dynamics represented by this simulation model, the transition between the high and low plateau regions is easily visible, with the blue and red circles in the figure indicating the beginning and end of discharge, respectively. The algorithm succeeds in detecting the dip point, with the detected location of the dip point marked by a black asterisk. As indicated in the figure, the algorithm declares that the dip point occurs 1,421.8 s after the onset of the simulation cycle. The transition point where cell voltage reaches a local minimum, in contrast, occurs 1,419.3 s after the onset of the simulation cycle. The difference between these two times represents a dip point detection error of 0.17% if one defines the dip point as occurring at the point of locally minimal voltage. This work foregoes this definition in favor of detecting the dip point based on the change in dV/dQ slope: a more robust approach that enables dip point detection even if voltage does not hit a local minimum between the high and low plateaus.

Three different experimental validation studies were performed, in addition to the above simulation study, using custom laboratory-fabricated Li–S cells. The first experimental study involved cell cycling at a relatively slow C-rate of approximately C/7, as shown in Fig. 5. The transition between the high and low plateaus manifests itself clearly as a dip point for this slow C-rate, and is successfully detected by the proposed algorithm. Once again, the difference between the moment in time declared by the algorithm as the dip point versus the moment in time when a local voltage minimum is reached is very small — in this case, reflecting a 1.32% detection error if one defines the dip point as the point of locally minimal voltage. This reaffirms the earlier, simulation-based validation of the dip point detection algorithm's performance in a quantitative manner, using experimental data. Moreover, as indicated on the figure, the root mean square error with which the proposed Taylor series expansion method fits the battery cell voltage for the time windows immediately preceding and following every candidate dip point, averaged over the entire test cycle, is 0.16 mV. This is an encouraging accuracy level that provides quantitative justification for this work's use of a simple first-order Taylor series expansion as a foundation for dip point detection.

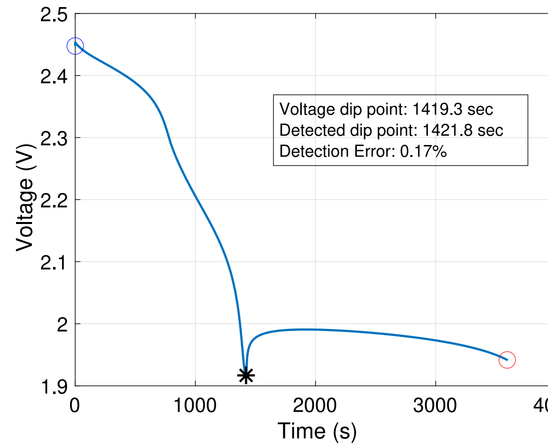

Fig. 4. Dip point detection result for simulated model of Li–S.

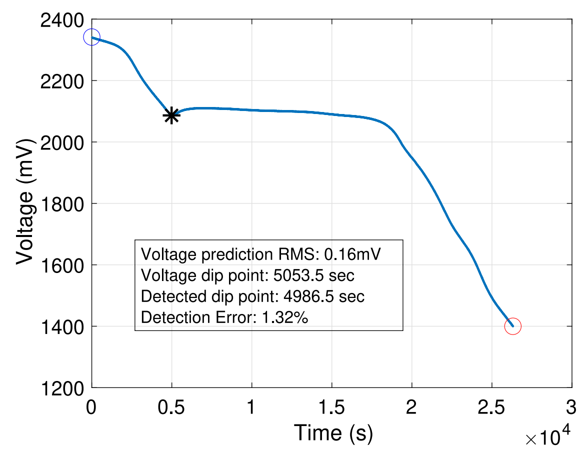

Fig. 5. Dip point detection result for slow current discharge Li–S cell.

The second experiment focused on evaluating the dip point detection algorithm for multiple charge/discharge cycles applied to the same Li–S cell. One goal of this validation study was to assess the repeatability of the algorithm. Another goal was to push the algorithm to its limit by cycling the Li–S cell at a C-rate high enough that a *transition* is evident, but not necessarily in the form of a *dip point*. This is important for providing a qualitative (as opposed to quantitative) validation of the robustness of the algorithm to different charge/discharge conditions, including conditions where the transition point does not manifest itself clearly as a dip point. Fig. 6 shows the dip point detection results for multiple cycles, with Fig. 7 zooming in to show the results for the first of these cycles. Each battery cycle clearly shows an initial rest period, constant-current charge period, a constant-voltage charge period, a rest period during which the battery cell self-discharges, a constant-current discharge period, and finally a rest period marked by battery voltage relaxation. The algorithm successfully detects the transitions between the high and low plateaus during all discharge cycles, even though the C-rate used is large enough that they do not manifest themselves as "dip points". This is a valuable feature of the algorithm, underscoring the robustness of the algorithm to different C-rates. Much of this robustness arises from the deliberate design of the algorithm to detect changes in dV/dQ slope, as opposed to the less robust definition of the dip point as a point of zero slope. In practice, the precise locations of the transition points detected by the algorithm depend on the widths of the data windows used for detection. The use of a first-order Taylor

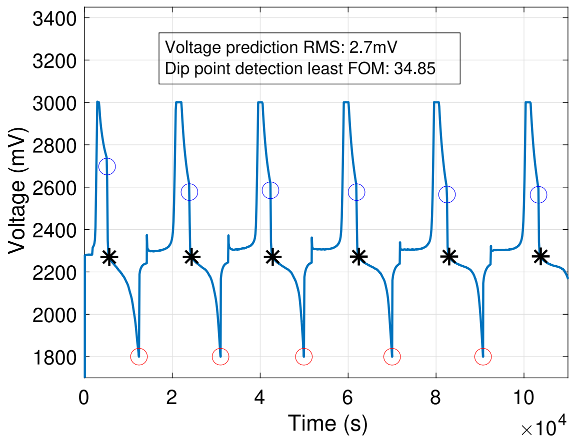

Fig. 6. Multi-cycle test results for a single Li–S coin cell.

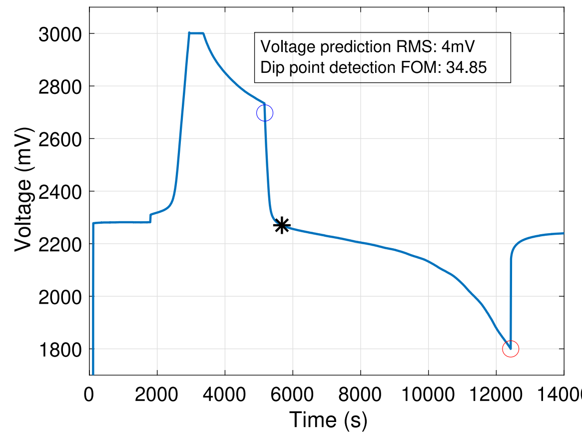

Fig. 7. Single cycle test results for a single Li–S coin cell.

series expansion as a foundation for dip point detection continues to be supported by the accuracy with which this expansion is able to capture battery terminal voltage fluctuations during the time windows preceding and following each candidate dip point. For the full set of 6 cell charge/discharge cycles, for example, the root mean square value of this voltage prediction error is 2.7 mV, and for the specific cycle shown in Fig. 7, it is 4 mV.

The accuracy with which this article's linear regression approach detects the large change in dV/dQ slope at the dip point is critical for the success of the algorithm. Therefore, it is possible to define a quantitative "figure of merit" that evaluates the algorithm by evaluating this accuracy. Specifically, suppose that the estimated change in dV/dQ slope at the dip point is $\hat{\delta\gamma}$, in Volts per Coulomb. Moreover, suppose that the standard errors associated with the use of linear least-squares estimation to determine the dV/dQ slopes immediately preceding and following the dip points are σ_v^p and σ_v^f, respectively (also in Volts per Coulomb). Then one can potentially define a dimensionless figure of merit in terms of the ratio of $\hat{\delta\gamma}$ to the maximum of σ_v^p and σ_v^f, as shown below. The use of the maximum of σ_v^p and σ_v^f in the denominator of this expression leads to a conservative FOM, and the FOM itself is intuitively a dimensionless signal-to-noise ratio associated with the estimation of the change in dV/dQ slope. One can compute this FOM for each transition point shown in Fig. 6. The smallest resulting FOM value corresponds to the cycle in Fig. 7, and has a value of 34.85. This conservative FOM value is, again, very encouraging in the sense that it highlights the success of the proposed regression approach in detecting the change in dV/dQ slope at the estimated dip point locations with very high accuracy. This, in turn, provides quantitative validation of the algorithm's success even in cases where there is no clearly visible "dip point" marking the transition between the high and low plateaus.

$$FOM = \frac{\hat{\delta\gamma}}{\max(\sigma_v^p, \sigma_v^f)} \tag{12}$$

The goal of the third experimental study was to validate the repeatability of the dip point detection algorithm across multiple battery cells. Four additional coin cells were fabricated and cycled in a manner similar to the cell in the second study. All cells were cycled using the same discharge current, but cell-to-cell variability caused this protocol to correspond to a slightly different effective C-rate for each cell. Each cell was cycled repeatedly to furnish multiple dip point detection results similar to Fig. 6. Fig. 8 shows the performance of the dip point detection algorithm for the first charge/discharge cycle applied to all five cells. The algorithm is successful in detecting the dip point in a consistent and repeatable manner across all five cells, even though the C-rates used in these experiments were sufficiently high to the point where the transition from the high to the low plateau does not manifest itself explicitly as a "dip point". Significant differences in capacity existed between these cells, as evident from the differences between their discharge curves. The proposed algorithm is consistently successful in dip point detection in the presence of this cell-to-cell variability.

5. Discussion, conclusions, and future work

The above results highlight the ability of the proposed algorithm to detect the passage of Li–S batteries through the transition between the high and low plateau regions of their discharge voltage curves. This has been achieved consistently, for multiple battery cells and multiple C-rates. Particularly exciting, in this context, is the algorithm's ability to detect the transition between the high and low plateaus even at high C-rates where this transition does not manifest itself explicitly as a "dip point". The algorithm's model-free nature is important for practical applications because it minimizes the effort and complexity associated with accurately modeling and parameterizing the discharge behavior of different Li–S cells. Moreover, the simplicity of the algorithm makes it straightforward for practical online implementation in battery management systems. The experimental validation results in this article focused on dip point detection during constant-current discharge. However, the algorithm can be easily extended to variable-current discharge scenarios by either subtracting the resistive term from Eq. (3) prior to regression or using regression to simultaneously estimate both γ and R. The experimental validation of such a simple extension of the algorithm is an important topic for potential future research, building on the results in this article. Another important potential extension of this work is the application of the algorithm to Li–S battery cells of different sizes and form factors, in order to ensure the algorithm's robustness to variations in these parameters.

The intent of the proposed dip point detection algorithm is not to replace SOC estimation for Li–S cells, but rather to provide an additional tool for Li–S battery management. This tool has multiple potential future benefits and extensions. For example, in a series string of Li–S cells, one could potentially construe the magnitude of cell-to-cell variability in passage through the dip point as an indication of the need for pack balancing. The idea is that those cells that pass through the dip point earlier during discharge are more likely to require charge replenishment through a balancing circuit. Another important potential future application of the proposed algorithm pertains to SOC estimation. In particular, previous work by the authors shows that unscented Kalman filtering (UKF) can be used for Li–S battery SOC estimation, with the caveat that the ideal dynamic model one should use

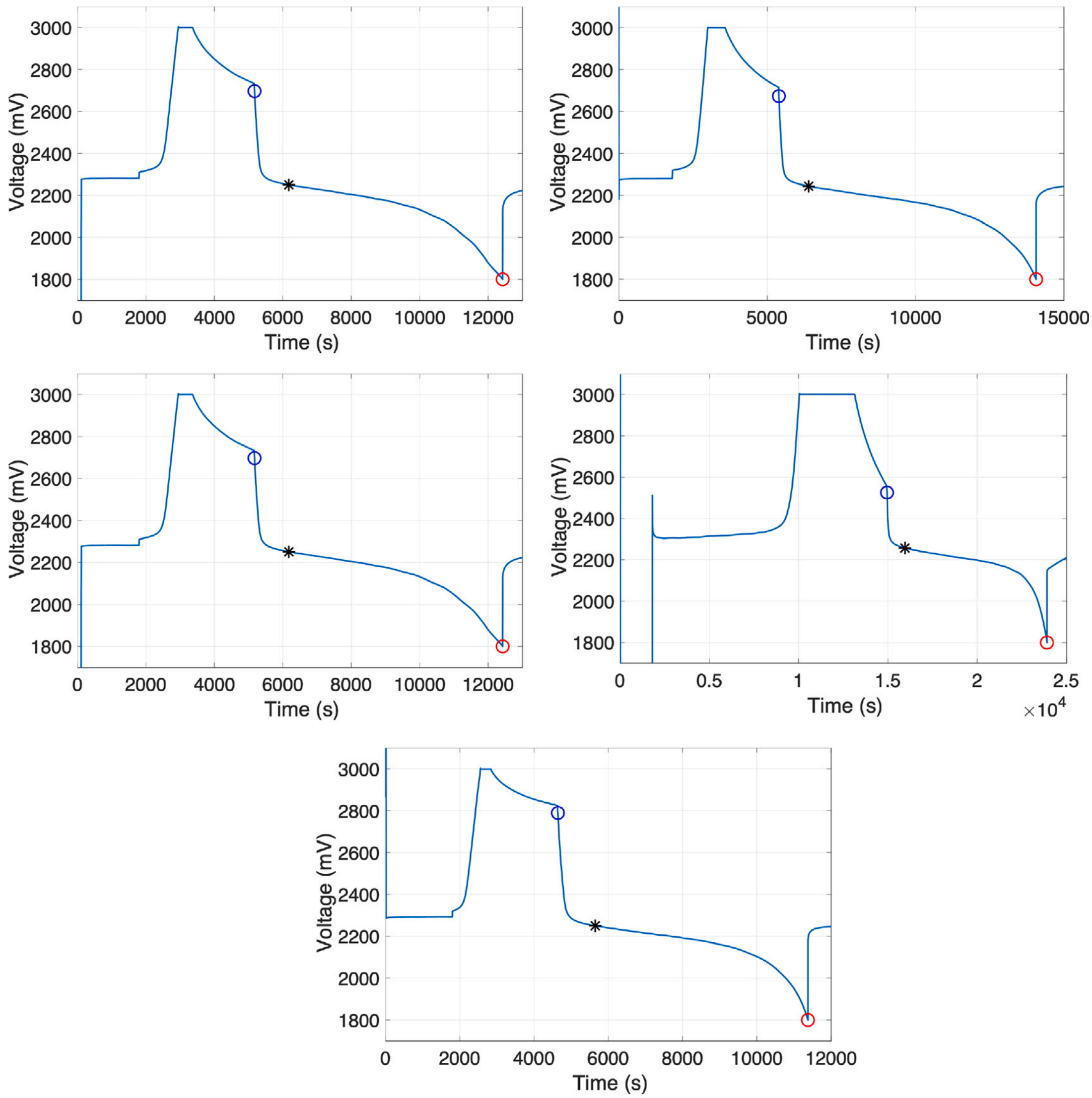

Fig. 8. Dip point detection results for multiple Li–S cells.

for such estimation is different for the high plateau versus low plateau region [23]. This creates an opportunity for using dip point detection as a foundation for the automated switching between different Li–S battery estimation algorithms optimized independently for the high and low plateau regions.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:Hosam Fathy reports financial support was provided by Maryland Energy Innovation Institute.

Data availability

Data will be made available on request.

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